

Basic Gene Sequence Analysis Using BLAST

Description

This project involved performing a basic nucleotide sequence analysis using the NCBI BLAST tool to identify similarities between an input DNA sequence and known sequences in biological databases.

Method

- Used NCBI BLAST (Nucleotide BLAST)
- Input a sample DNA sequence
- Analyzed alignment results, including percent identity, E-value, and query coverage

Results

The BLAST analysis showed multiple sequence alignments with different organisms, including *Sander lucioperca* and *Kluyveromyces marxianus*.

The highest similarity reached approximately 97% identity, with E-values indicating statistically significant matches.

Due to the short length of the input sequence, the results showed partial alignments across different species.

BLAST Output

Descriptions									
Sequences producing significant alignments									
Download Select columns Show 100 ?									
select all 5 sequences selected									
GenBank Graphics Distance tree of results MSA Viewer									
	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	PREDICTED: Sander lucioperca protein LYRIC-like (LOC116066574)_transcript variant...	Sander lucio...	60.2	60.2	55%	2e-05	97.14%	2857	XM_031322730.2
☒	PREDICTED: Sander lucioperca protein LYRIC-like (LOC116066574)_transcript variant...	Sander lucio...	60.2	60.2	55%	2e-05	97.14%	2925	XM_031322728.2
☒	Kluyveromyces marxianus strain CCT 7735 (UFV-3) chromosome 3 sequence	Kluyveromyc...	60.2	60.2	59%	2e-05	94.87%	1586801	CP009305.2
☒	PREDICTED: Zeugodacus cucurbitae uncharacterized LOC105216917 (LOC105216917...	Zeugodacus...	52.8	52.8	53%	0.004	94.29%	3008	XM_011191670.3
☒	PREDICTED: Pagrus major tudor domain containing 12 (tldr12)_mRNA	Pagrus major	52.8	52.8	44%	0.004	100.00%	4835	XM_073471363.1

Figure 1: BLAST alignment results showing sequence similarity across different organisms

Interpretation

The results suggest that the input sequence shares partial similarity with known biological sequences. The low E-values indicate that the matches are unlikely to have occurred by chance, while the moderate query coverage reflects alignment with only part of the sequence.

Skills Demonstrated

- Sequence analysis using BLAST
- Interpretation of bioinformatics results
- Use of biological databases (NCBI)